

Problem 1: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Play from 4 counters twice, swapping who starts. Draw or write the number taken at each turn. Circle the last move.

Game 1 moves:

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Game 2 moves:

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Problem 2: Play from 5 counters twice, swapping who starts. Record the moves and circle the last move. Use the same 1-or-2 rule.

Game 1 moves:

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Game 2 moves:

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Problem 3: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Start with 3 counters. Let the other player start. Try each of their first moves below. Record how many you take next.

Starting pile	Other player takes	I take next	Counters left
3	1		
3	2		

Problem 4: Start with 6 counters. Make two groups of 3. Let the other player start. Try each first move below and choose your reply so that exactly 3 counters remain. Then finish each game.

Other player takes	I take next	Counters left
1		
2		

Draw the two groups before the game:

Problem 5: Two players alternate taking 1 or 2 counters. Never take more than remain. Taking the last counter wins. Try 7, 8, and 9 counters. For each pile, choose whether to go first or second. If you go first, record your first move. Play twice from each pile and check your choice.

Pile	I choose first or second	My first move if I start
7		
8		
9		

Problem 6: Build 12 counters as four groups of 3. Let the other player start. After each of their moves, take enough to remove 3 counters between your two turns. Record the numbers left after each of your turns. Play until the counters are gone.

Draw my four groups:

Counters left after my turns: